Exploratory Analysis of UK Food Incident Notifications (2023–2024): Visual Analytics for Regulatory Risk Identification

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Abstract

Food incident notification systems form a critical component of the United Kingdom's food safety regulatory framework, supporting the identification of hazards and enabling timely regulatory intervention. However, without appropriate analytical and visualisation approaches, the large volumes of incident data generated by these systems can be difficult to interpret—limiting their usefulness for operational and strategic decision-making. This paper draws on UK food incident notifications from 2023–2024 to explore high-burden product categories, temporal patterns, dominant incident types, and reporting characteristics that can inform risk-based monitoring and compliance activity. A structured data science lifecycle, aligned with the CRISP-DM framework, was applied. Two open datasets published via data.gov.uk were merged and processed using Python (pandas), including standardisation of variables, date transformation, removal of invalid records, and deduplication. Visual analytics were primarily conducted in Power BI to support exploratory analysis and stakeholder-focused communication. To assess analytical tools and ensure reproducibility, equivalent static visualisations were also generated using Python (matplotlib). All visualisations avoided pie charts and included clear labelling and titles to enhance interpretability. The analysis reveals that food incident notifications are concentrated within a small number of product categories—particularly meat and meat products—highlighting clear candidates for focused regulatory oversight. Temporal analysis shows fluctuations in monthly incident volumes, suggesting potential seasonal or operational risk windows. The composition of incident types points to the continued prominence of microbiological and labelling-related issues, while variations in notifier and source patterns indicate potential reporting bias. Together, these findings illustrate how visual analytics can translate regulatory datasets into actionable insight, while also reinforcing the importance of cautious interpretation due to data quality limitations. Overall, this study demonstrates that descriptive visual analytics—applied within a structured analytical lifecycle and informed by critical evaluation of data limitations—can offer meaningful decision support for food safety regulation, without relying on predictive modelling.

Use of Generative AI

This assignment utilises Generative AI in the processes of brainstorming, planning, and editing. Evidence is provided in the appendix.

Key Words:

Food safety analytics, data from regulators, visual analytics, and incident monitoring

1. INTRODUCTION

Food safety surveillance is a fundamental component of public health protection and regulatory oversight in the United Kingdom. Incident notification systems provide a structured mechanism for capturing data on hazards associated with food and feed, including microbiological contamination, chemical risks, allergen mislabelling, and failures in production or distribution controls. When analysed effectively, these datasets can support evidence-based decision-making by enabling regulators and industry stakeholders to prioritise inspections, allocate resources efficiently, and implement targeted compliance interventions.

However, despite their value, food incident notification datasets present significant analytical challenges. They are typically large and categorical in nature, and are influenced by variability in reporting practices, inspection intensity, and detection capability. As a result, meaningful insights cannot be derived through simple tabulation alone. Visual analytics offers a robust solution by allowing analysts to explore complex datasets, identify patterns, trends, and anomalies, and communicate findings in an accessible format—particularly for non-technical audiences.

This paper presents an exploratory visual analysis of UK food incident notifications from 2023 to 2024. Applying a structured data science methodology based on the CRISP-DM framework, the study investigates temporal trends, product category concentration, incident type composition, reporting patterns, and available geographic indicators. The aim is not to predict future incidents, but to generate insight that can highlight areas of regulatory concern and demonstrate the value of visual analytics in supporting operational and strategic decision-making within a regulatory context.

The study is guided by the following research questions:

- Which food product categories are most frequently associated with reported incidents?
- Are there identifiable temporal or seasonal trends in food incident notifications?
- Which incident types dominate reporting over the two years?
- What reporting patterns and potential biases are evident within the dataset?

2. Background Reading and Research

Food safety surveillance systems play a central role in safeguarding public health and supporting regulatory oversight in the United Kingdom. Incident notification mechanisms are designed to systematically capture information about food and feed-related hazards—including microbiological contamination, chemical risks, allergen mislabelling, and failures in production or distribution controls. These systems are essential not only for immediate risk response, but also for long-term trend analysis that informs regulatory prioritisation and policy development [2], [3].

However, prior research has shown that regulatory incident datasets are inherently observational in nature, shaped by factors such as inspection activity, reporting behaviours, and enforcement priorities [10]. As a result, incident counts should not be interpreted as direct indicators of underlying food safety risk. Rather, they reflect a combination of actual hazard occurrence and the processes used to detect and report those hazards. This underscores the importance of applying descriptive and exploratory analytics—focused on identifying patterns, concentrations, and temporal changes—rather than attempting causal inference or predictive modelling [9].

Visual analytics has become a widely used approach in public-sector and regulatory contexts to manage the complexity of such datasets. As Keim et al. [5] suggest, visual analytics facilitates sense-making by integrating computational analysis with human judgment, enabling users to navigate complex, uncertain datasets while maintaining transparency. In regulatory environments, visualisations are especially valuable for highlighting trends, category concentrations, and anomalies that may require further investigation—all while remaining accessible to non-technical decision-makers.

In the domain of food safety, earlier studies have shown that analysing incident notifications by product category and hazard type can support risk-based regulatory approaches [2], [11]. At the European level, surveillance reports consistently find that a small number of product categories account for a large share of reported incidents. This points to the potential effectiveness of targeted monitoring strategies compared to blanket inspection policies [2]. Additionally, time-based analyses often reveal seasonal trends aligned with production cycles, import patterns, and inspection schedules—further reinforcing the value of time-series visualisation for regulatory insights [3].

Despite their utility, food incident datasets face ongoing data quality challenges. Issues such as inconsistent categorisation, missing origin data, and the lack of severity or outcome measures hinder the comparability and interpretability of records [9]. The UK Office for National Statistics stresses the importance of explicitly acknowledging such limitations when working with administrative and regulatory data and recommends maintaining transparency about uncertainty as a key element of responsible analysis [9].

Building on this foundation, the present study applies a structured CRISP-DM lifecycle [1] and adopts visual analytics as its primary analytical methodology. This approach was selected to ensure interpretability, transparency, and decision relevance when analysing regulatory incident data. By combining multiple visual perspectives—including temporal trends, product category concentration, incident type breakdowns, reporting patterns, and geographic indicators—the study aims to generate actionable insights for stakeholders while critically reflecting on the limitations of the dataset itself.

3. Methodology

3.1) Data Science Process (CRISP-DM): This study utilises the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to ensure a structured, transparent, and reproducible analytical workflow. CRISP-DM was chosen due to its widespread use in applied data analytics and its effectiveness in aligning organisational objectives with analytical outcomes. The framework consists of six phases: Business Understanding, Data Understanding, Data Preparation, Analysis, Evaluation, and Deployment. These phases provide a coherent and iterative structure to guide the transformation of raw incident notification data into insights that are relevant for decision-making.

3.2) Business Understanding: Food incident notification systems are a cornerstone of the UK's food safety regulatory framework, playing a key role in identifying hazards, guiding inspection priorities, and protecting public health. However, the datasets generated by these systems are typically large, categorical, and observational in nature—making it challenging to extract meaningful, actionable insights without the use of appropriate analytical methods.

The primary objective of this study is to uncover patterns within reported food incidents that can support risk-based regulatory decision-making. In particular, the analysis is designed to inform:

- the prioritisation of inspection and monitoring activities,
- targeted compliance efforts focused on high-burden product categories, and
- a better understanding of temporal trends and reporting behaviours.

Rather than attempting to predict future incidents, the study adopts a descriptive and exploratory visual analytics approach to answer practical, operational questions—such as where incidents are most concentrated, how volumes fluctuate over time, and which types of issues are reported most often.

3.3) Data Understanding: The dataset used for the study encompasses UK food incident notifications published through the UK Government open data platform (https://www.data.gov.uk/dataset/8baf7984-2a90-45ba-

a497-8dff82c0443a/received-food-incident-notifications) for the years 2023 and 2024. Two annual datasets were combined into one analytical dataset comprising 3,587 unique incident records.

Key variables are:

- Incident identifier,
- Date received,
- Incident type,
- Product category,
- Notifier type,
- Incident source,
- Country of origin, where available.

These variables support time-series analysis, categorical comparison, and limited geographic exploration. Initial cleaning and exploration of the data suggested that many of the problems associated with regulatory datasets were present, including missing values, inconsistent categorisation, and variation in reporting across notifier types.

3.4) DATA PREPARATION: Data preparation was conducted using Python (pandas) to ensure transparency and reproducibility throughout the analytical workflow. The focus was on enhancing analytical validity rather than applying exhaustive transformations. To support consistent categorisation and aggregation, column names were standardised across the annual datasets before merging. Date variables were converted to a uniform format and retained at monthly granularity, enabling time-series analysis aligned with the study's objectives.

To ensure accuracy in reported counts, duplicate records were removed using the unique incident identifier. In addition, invalid or placeholder values—such as "Unknown" or null entries—were excluded where they could distort frequency distributions or temporal patterns. These steps helped ensure that the final dataset reflected unique, verifiable incidents and that comparisons across time periods and product categories remained meaningful.

The cleaned dataset was then exported as a single CSV file, ready for downstream visual analytics in Power BI and Python-based tools.

3.5) VISUAL ANALYTICS AND TOOLS:

3.5.1. POWER BI (PRIMARY VISUALISATION PLATFORM): Power BI was selected as the primary visual analytics platform because it was most suitable for organisational and regulatory stakeholders. Its support for rapid development of interactive dashboards, with clear labelling and intuitive filtering, makes it valuable for exploratory analyses and communication of findings to non-technical audiences.

The following were created using Power BI in this study:

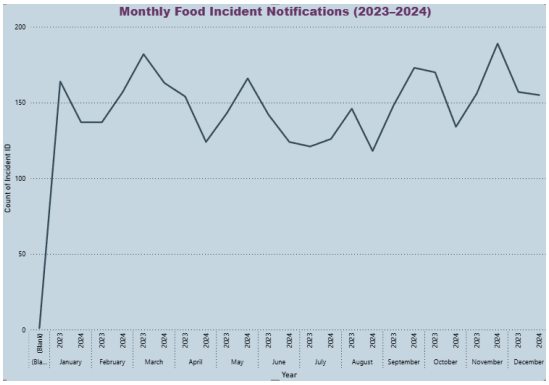


Figure 1: Monthly Food Incident Notifications (2023–2024)

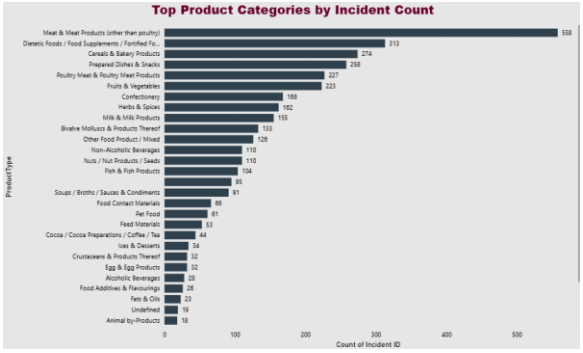


Figure 2: Top Product Categories by Incident Count

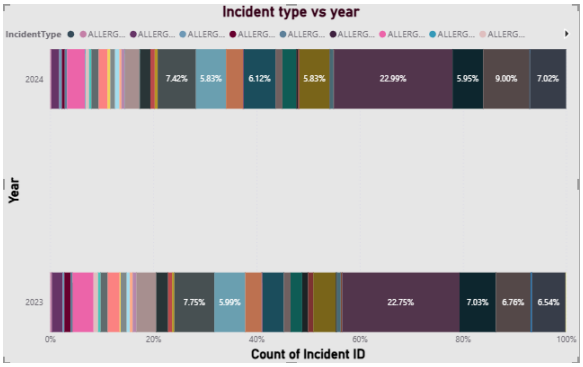


Figure 3: Incident Type vs Year

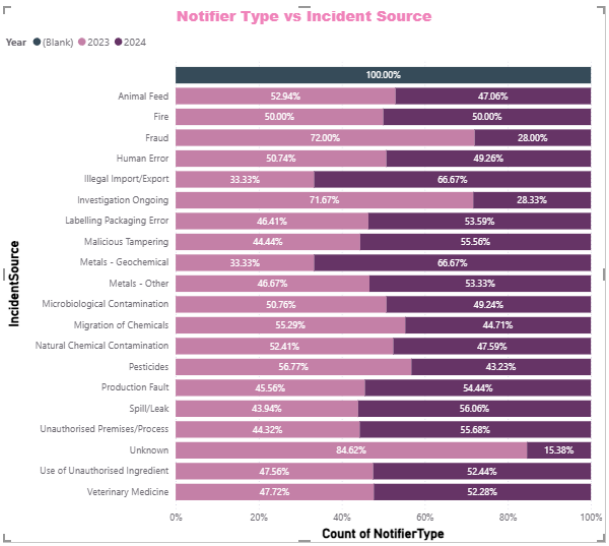


Figure 4: Notifier Type vs Incident Source

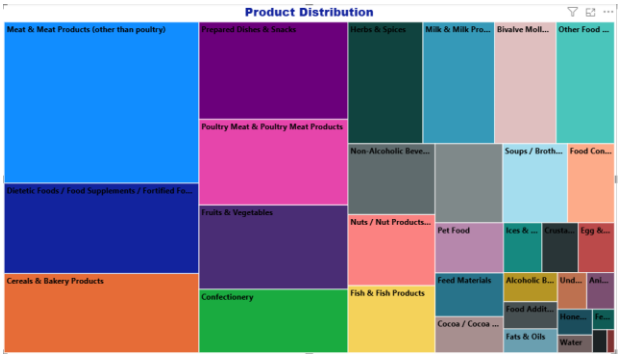


Figure 5: Product Distribution (Tree map)

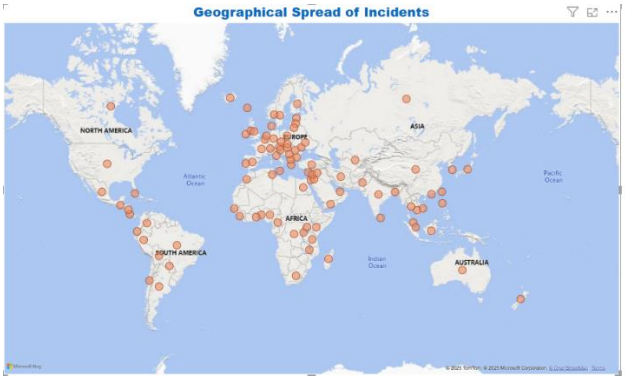


Figure 6: Geographical Spread of Incidents

The most significant limitation of Power BI is that complicated transformations and calculations are less transparent to readers without access to the report file, and reproducibility depends on the correct versioning of the report.

3.5.2. PYTHON (PANDAS AND MATPLOTLIB) AS A COMPARATIVE PLATFORM: Static visualisations of equivalents were created, using Python—more specifically, pandas and matplotlib—for evaluation and demonstration of reproducibility of the analytical tools. Python was chosen as a low-level comparative platform because the language is in very active use in data science, and its output is fully reproducible via code.

Python was used to recreate the key visuals, including:

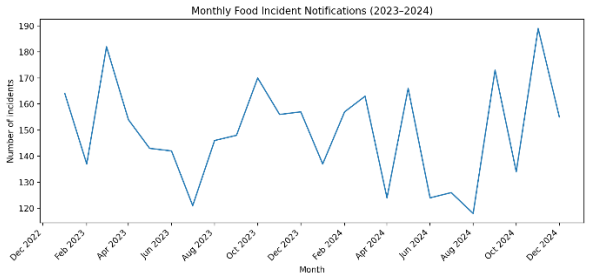


Figure 7: Monthly incident trends

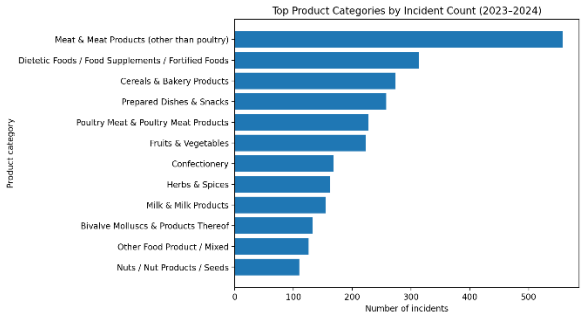


Figure 8: Top Product Categories by Incident Count

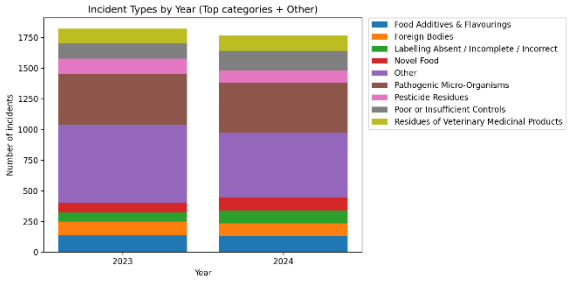


Figure 9: Stacked incident types by year

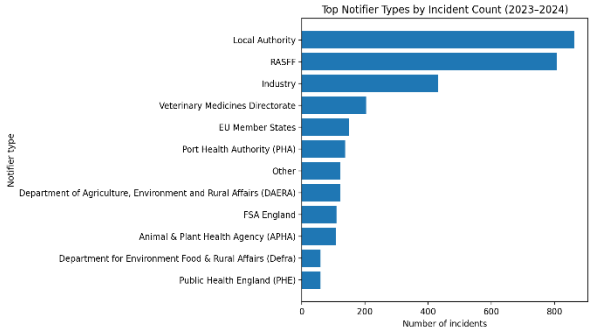


Figure 10: Notifier-type distributions

The strengths include transparency of transformations, ease of version control, and reproducibility of the results. The Python-

based visualisations are generally less interactive and less accessible to the non-technical stakeholder compared with the dashboards made in Power BI.

3.5.3. Tool Evaluation Summary: In this respect, the use of both Power BI and Python emphasises the need to choose appropriate analytical tools according to audience and purpose. Power BI is great for stakeholder engagement, operational dashboards, and speedy exploratory analysis, whereas Python enables reproducible and auditable workflows that enhance methodological rigour. By using both tools together, one can make sure that sound analysis is performed while also ensuring that findings can be clearly communicated to decision-makers.

3.6) Ethical and Analytical Considerations: The dataset is sourced from open government sources, and no personal or sensitive identifiers are included. Ethical considerations then revolve around responsible interpretation rather than data privacy. Incident counts may reflect reporting intensity, inspection activities, or notifier behaviours rather than true underlying risk. Results are therefore provided at a high aggregate level, and conclusions avoid attributing causality or responsibility to an organisation or country.

4. Discussion and Findings

It is important to recognise that food incident notifications represent reported events rather than confirmed cases of harm. As such, the dataset reflects not only underlying food safety risks but also the dynamics of surveillance, inspection, and reporting systems. Variations in inspection intensity, notifier behaviour, and reporting practices can all influence the volume and distribution of recorded incidents. Given these factors, the analysis presented in this study focuses on the concept of reported burden—that is, patterns within the data as captured through official notification processes. No claims are made about true prevalence or causality. Instead, the aim is to identify areas of potential concern and highlight trends that may warrant further investigation. This distinction is critical for the responsible use and interpretation of regulatory incident data, ensuring that findings are contextualised within the limitations of the reporting system itself.

Figure 2: Monthly Food Incident Notifications (2023–2024): It shows the volume of food incident notifications received monthly over the two years. Incident counts are variable, rather than fixed, with several peaks and troughs readily identifiable for different months. This pattern may indicate that incident notifications are sensitive to temporal factors, such as seasonal production cycles, import patterns, and/or temporal variations in inspection or reporting activity. The fact that there is no consistent trend would also confirm that the yearly distribution of food safety risks and/or detection efforts is not constant.

Operational implications: However, operationally speaking, the months showing consistently higher incident volumes could serve as risk windows. This would let regulators plan much-needed inspection campaigns or increase their capacity for monitoring during those periods. Yet again, these kinds of patterns need not be interpreted as direct evidence of increased underlying risks since peaks may also be an indication of heightened surveillance or reporting.

Figure 2: Top Product Categories by Incident Count: Incident notifications are primarily concentrated in a limited number of product categories, as illustrated in Figure 2. The category with the highest number of incidents is meat and meat products, followed by dietetic or fortified foods, cereals and bakery products, and prepared dishes and snacks. This concentration indicates that a small number of product categories account for a disproportionately large share of the reported incidents. This clustering aligns with the complexity of certain supply chains and the increased regulatory scrutiny applied to higher-risk products.

Operational implications: This finding forms the basis for a risk-based approach to regulatory oversight. Targeted inspection, compliance audit, and industry guidance are best leveraged for high-burden categories for maximum impact. Low-frequency categories should not be disregarded; smaller volumes may reflect under-reporting rather than absence of risk.

Figure 3: Incident Type vs Year: It shows the proportional distribution of incident types between 2023 and 2024. Although there are a few dominant classes in both, some changes in composition may indicate an underlying shift in the relative contribution across hazard or issue categories. These shifts could indicate changing risk profiles, a shift in the focus of regulators, or improvements in the detection and reporting practices. For instance, an increased proportion of labelling-related incidents may point to increased enforcement activity or increased awareness of allergen controls.

Operational implications: Understanding the changes in incident type composition also allows regulators to revise guidance and enforcement priorities. Where specific issue types are increasing, focused interventions—such as increased labelling checks or process audits—may be appropriate. However, interpretation is limited by the lack of information on incident severity and possible changes in categorisation practice over time.

Figure 4: Notifier Type vs Incident Source: Figure 4 illustrates the differences in reporting patterns between incident sources and years. Some incident types seem to be more related to reporting channels, which may suggest that reporting is not uniform for different types of notifiers. This would suggest that incident distributions found may be influenced by the reporters of incidents and the mechanisms of reporting, rather than the hazards themselves.

Operational implications: To decision-makers, this finding reinforces the need for caution with incident counts as a proxy for

risk in any quantitative analysis. Any biases in reporting might be over- or under-representing certain issues. Incident data integrated with other evidence sources, such as inspection outcomes or recall data, will support greater confidence in prioritisation decisions.

Figure 5: Product Distribution (Tree map): It presents the product category distribution summarised by a tree map that emphasises the dominance of a few categories while still providing the representation of the extent of products implicated in incidents. Tree maps are good at making relative scale apparent immediately, but they are not as sensitive for comparing mid-ranking categories as bar charts.

Operational implications: This is a useful visual to communicate at a high level to stakeholders, for example, in presentations and dashboards. However, for operational prioritisation, ranked bar charts remain more suitable. The tree map, therefore, complements rather than replaces other visualisations.

Figure 6: Geographical Spread of Incidents: This maps the geographic origin of incidents where origin country data is available. The incidents are distributed across multiple regions, indicating that both domestic and imported supply chains contribute to reported incidents. The presence of missing or 'unknown' origin values limits the reliability of geographic interpretation.

Operational implications: Geographic analysis can inform targeted import controls and international collaboration. However, incomplete origin data preclude conclusions from being more than tentative. Improvement in the completeness and consistency of origin reporting would greatly enhance the value of geographic analytics in future work.

5. Data Analytics Challenges and Solutions

5.1. **Key Challenges:** Several analytical challenges have been identified. First, there is a clear presence of reporting bias, as the volume and type of incidents are influenced by both the behaviours of the notifiers and the mechanisms used for reporting. Second, inconsistent categorisation and missing values—especially regarding the origin country—reduce the comparability of records. Third, the dataset lacks measures of severity or outcome, which implies that all incidents are treated as having equal weight, regardless of their impact.

These challenges indicate that incident counts reflect the reported burden rather than the true prevalence or risk.

5.2. **Proposed Solutions:** Improvement opportunities to further enhance analytics and decision support in the future include:

- Standardisation of taxonomies and data dictionaries to decrease inconsistency in classification.
- Using appropriate validation rules at data entry to minimise missing or ambiguous values.
- Introduce severity or risk-weighting indicators, where possible, to differentiate high-impact incidents.
- Combination of incident data with inspection, recall, and enforcement datasets to enable triangulation.
- Using dashboard-based early-warning systems to support proactive monitoring during periods of high activity.

6. Conclusions

This study adopted a CRISP-DM-compliant visual analytics methodology on UK food incident notifications from the year 2023–2024. This analysis reveals that incidents are focused on a few categories of products and show temporal variations across different months. The changes in the composition of the incident types and reporting patterns suggest more complexity when interpreting datasets under regulation.

These findings therefore support risk-based prioritisation of inspections and focused compliance interventions from an organisational perspective. However, data quality limitations and reporting bias mean that incident counts should be interpreted with caution. The reliability and value of future analytics will be significantly enhanced by strengthening data standardisation and completeness and integrating these with outcome measures.

Overall, the study shows how visual analytics can transform regulatory incident data into actionable insight while maintaining transparency about limitations and uncertainty. Such insights are especially valuable in resource-constrained regulatory settings, where evidence-based prioritisation is critical to maximising the impact of inspection and enforcement efforts.

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APPENDIX

As per the module guidelines on ethical use of AI, this Appendix provides proof in the form of evidence about the use of generative artificial intelligence (AI) tools for the preparation of this assignment. Generative AI played a role in effective brainstorming, planning, and editing. The student performed and verified all analytical decisions, data preparation, visualisations, writing, and interpretations.

1. Evidence of AI Interactions

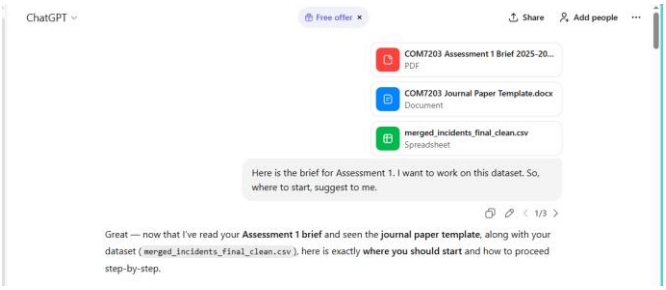


Figure A1: Screenshot of brainstorming and planning

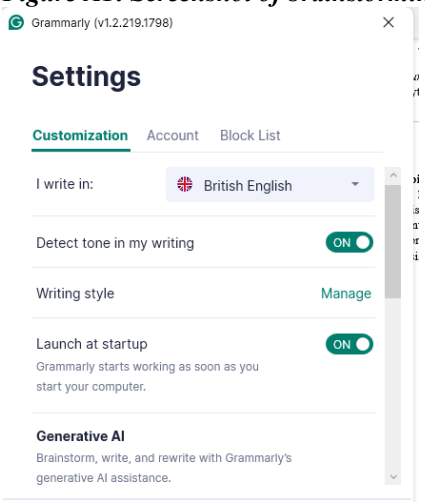


Figure A2: Screenshot of editing.

2. Statement of Academic Responsibility

The student confirms to the best of her knowledge that all analytical decisions, interpretations of visualisations, conclusions, and recommendations in this assignment are solely those of the student. The use of generative AI was for supporting the analysis, not replacing certain independent analysis or critical thinking.